

系統程式設計

Lab8 Race condition

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1. How did you implement the processes?

I called TELL_WAIT() before forking to initialize the 2 sync pipes, then a for loop to call fork() 5 times, creating 1 child per year. Each child's index I determines which year's data files it handles. After creating all 5 child processes the parent goes on past the loop while each child enters their own block. The parent then waits for all the child processes using wait(NULL) in a loop.

2. How did you update the file "sum.txt"?

Each child process reads the assigned weekly file and for each of the 7 days in the week, it sums all the 96 values into d_sum, then calls accumulation(d_sum).

In accumulation(), sum.txt is opened with r+ mode, using fscanf() to read current total, then adding d_sum to it, then overwriting old value with new one.

3. How did you solve the race condition?

Before each child calls accumulation(), they'll be blocked by WAIT_PARENT(), which reads on the pipe until the parent sends a signal. After the child finishes calling accumulation(), the child calls TELL_PARENT(getppid()) to write back to the parent. The parent loops through all five child processes calling TELL_CHILD(pid[i]) to unblock 1 child process at a time and then using WAIT_CHILD() to wait for the child to finish. This way it guarantees that at most 1 child process will be calling accumulation() hence eliminating the race condition.

4. What is the average daily power consumption over the 5 years? (daily means for one day)

Please attach screenshot !

Day_Average = 4743

```
lk@LK Lab8 % make
gcc lab8.c tellwait.c error.c -o run
lk@LK Lab8 % ./run
Day_Average = 4743
lk@LK Lab8 %
```